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Secure Satellite Communication: Performance and cryptographic analysis of chaotic injected AES resistant to side-channel attacks targeting the S-box via machine unlearning

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Simone Descontus
ID number 2116496

Advisor
Prof. Emiliano Casalicchio

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Sapienza University of Rome

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Author's email: simo.desk04@gmail.com

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Chapter 1

Introduction

1.1 Thesis Objective

Inspired by the foundational work of Kumar et al. [1], the primary objective of this thesis is to develop a proof-of-concept implementation of *AES with chaotic injection*. This is achieved by extending the open-source `tiny-aes-c` library [2] and adapting it to comply with modern satellite communication standards through the integration of the *AES-256-GCM* mode of operation [3].

Furthermore, this study aims to analyze the integration of various chaotic maps within the cipher's architecture. These configurations will be rigorously benchmarked to evaluate execution time and energy consumption, thereby identifying the optimal solution for the resource-constrained SATCOM environment. Concurrently, statistical and cryptographic tests will verify that altering the standard encryption algorithm does not degrade its fundamental cryptographic properties.

Building upon these analyses, the thesis proposes an architectural optimization tailored for satellite communications. A comprehensive comparison will be provided between the baseline chaotic algorithm proposed by Kumar et al. and a modified iteration engineered to yield superior performance metrics, rendering it highly appropriate for space applications.

Lastly, significantly diverging from the scope of Kumar et al. [1], this research will simulate a Machine Learning-based Side-Channel Attack (SCA) aiming for complete key recovery. This simulation serves a critical dual purpose: it demonstrates a devastating threat vector that is entirely feasible against standard AES-256-GCM in current SATCOM architectures, while simultaneously proving that the proposed chaotic implementation successfully immunizes the system against such advanced profiling attacks.

Chapter 2

Background and State-of-the-Art of Satellite Communications

Satellite communications (SATCOM) are vastly different from how they appeared just a few decades ago. At that time, space-based communications were mainly represented by government activities, characterized by larger and more expensive satellites than the ones used today. This era is also known as the *old space* [4, p. 377], a term often used to describe the space applications that emerged during the Cold War.

On the other hand, since the beginning of the *new space* era [4, p. 377–398], thanks to the ever-decreasing cost of satellite launches and the enormous technological progress that has been made in recent years, space has become increasingly populated by satellites used for commercial purposes, such as:

- *Telecommunications and broadband connectivity*: they help to provide an internet connection even in the most remote regions or those that are hard to reach with a physical connection [4, p. 373–381].
- *Television broadcast and multimedia streaming*: due to the constant demand for high-quality multimedia, the wide coverage of satellite communications helps to deliver content to a large number of users more efficiently [4, p. 373–381].
- *Global Positioning System (GPS)*: numerous sectors make use of satellite-based geolocation to plan routes or monitor the movement of resources. In addition to this, a large number of people use geolocation daily to plan their travels [4, p. 378–381].
- *Earth observation*: thanks to more affordable satellites, it is now possible for private entities to start their own Earth observation programs for scientific purposes, such as environmental analysis. Moreover, satellites observing the Earth are widely used for meteorological purposes [4, p. 372–381].
- *Internet of Things (IoT)*: because of the multifaceted nature of IoT devices, which can take on many different forms, from smart homes to means of transport, it is essential that the coverage area serving these devices is as wide as possible. Such needs are effectively addressed by using SATCOMs as the main communication method [4, p. 372].
- *Satellite as a service (SataaS)*: while it is true that the cost of satellite lounches is significantly decreased, it remains unaffordable for many private companies that would otherwise

benefit of the wide coverage provided by SATCOMs.

To address this challange, majour caerospace operators can lease resources on their satellites to third-party users, granting them access to space infrastructure without the need for additional launches [5].

Along with the opportunities offered by the *new space*, the critical requirements of the *old space* have not diminished. Today, satellite communications remain essential for handling mission-critical data exchanges, such as those required by aviation and maritime operations [4, p. 397–398], as well as for the coordination of rescue services in disaster or crisis situations [4, p. 388–389].

Given the critical nature of the communications handled by SATCOM systems, it is becoming increasingly important to protect the information shared within these networks from malicious actors. To fully understand the scale of these security challenges and how threats can compromise the network, it is essential to analyze its underlying infrastructure. A modern SATCOM system is not a monolithic entity, but rather a complex, distributed architecture. Breaking down this infrastructure into its fundamental segments is a crucial step to accurately identify the specific attack surfaces that adversaries might exploit.

2.1 SATCOM Architecture

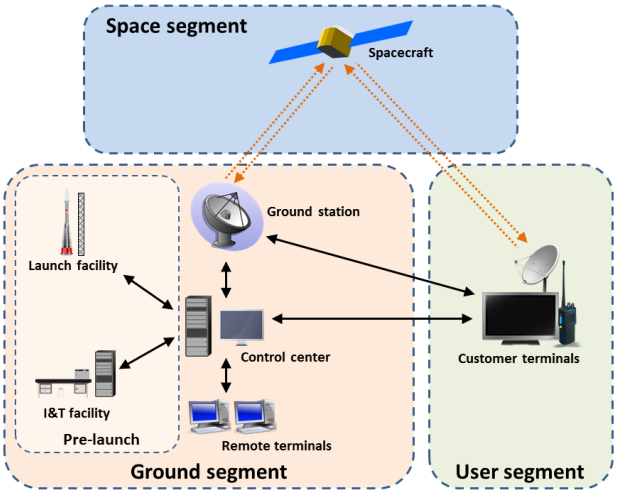


Figure 2.1:
Architectural overview of space, ground, and user segments in satellite systems [6].

Satellite communications are typically classified into three fundamental segments, although the exact categorization may vary across the literature. One common classification framework, as described in extended SATCOM state-of-the-art analysis provided by Salim et al. [4], divides the architecture into the *space segment*, the *ground segment*, and the *link segment*. In this model, the *ground Segment* encompasses both the *ground Stations*, which act as gateways for devices lacking direct satellite communication capabilities, and the *user segment*, which consists of end-user terminals equipped for space links. Another classification model still includes the *space segment* and

the *ground segment*, but explicitly separates the *user segment*, treating the *link segment* merely as a passive communication channel [7, p. 5671].

While all these components are crucial to maintain the network's operability, this thesis will adopt the classification proposed in [4]. Grouping the ground stations and end-user devices under a unified *ground segment* is conceptually sound for terrestrial entities. More importantly, treating the *link segment* as a primary, distinct architectural component is essential for the scope of this study, as the physical transmission channel introduces specific vulnerabilities and security challenges that are central to the analyzed threat model.

2.1.1 Space Segment

The space segment is represented by the satellites in orbit, each composed of the *bus*, which handles onboard systems such as power and thermal management, and the *payload*, which is the part of the spacecraft designed to fulfill the satellite's specific mission [4, p. 381–383]. Examples of payloads include telescopes to analyze distant objects or sensors to measure spatial properties.

Depending on the satellite's mission, its orbital height from Earth can vary radically [7]:

- *Low Earth Orbit (LEO)*: due to their low altitude, usually between 500 and 1200 km, these types of satellites are ideal for extending terrestrial networks thanks to their low latency (around 30 ms). However, because they are so close to the Earth, it is often necessary to deploy a group of interconnected *LEO* satellites, known as a *satellite constellation*, in order to cover a wider area of the planet. Additionally, because of the high risk of collision with space debris, the lifetime of *LEO* spacecraft is around 5 to 7 years, which is usually shorter than that of other satellites.

- *Medium Earth Orbit (MEO)*: usually positioned at an altitude of 2000 to 20000 km, they represent a midpoint between coverage and latency (around 125 ms). A significant share of *MEO* spacecraft function as GPS satellites, and typically 6 units are necessary in order to provide adequate service. Because of their intermediate position, *MEO* satellites often serve as relay nodes between *LEO* and *GEO* satellites. Finally, their operational lifetime is estimated at 10 to 15 years, depending on their altitude and exposure to radiation.

- *Geostationary Equatorial Orbit (GEO)*: operating in the highest orbital layer at approximately 36000 km, these spacecraft take exactly 24 hours to complete an orbit, appearing stationary to an observer on Earth. Thanks to their high altitude, they can cover one-third of the entire planet's surface, making them ideal for broadcasting. However, it is important to account for the latency accumulated during transmission (around 600 ms). Due to their distance from the planet and the large quantity of resources needed to launch *GEO* spacecraft, they are usually built to last longer, typically around 15 to 20 years.

2.1.2 Ground Segment

This segment represents the terrestrial infrastructure that communicates with satellites. Depending on the role of the receiver or sender, it can be classified into several sub-components, the primary ones being [4, p. 383–384]:

- *Ground Stations (GS)*: equipped with dedicated hardware, they can transmit and receive signals even from the most distant satellites. A vast amount of traffic passes through them, as the majority of terrestrial devices lack the necessary peripherals to communicate directly with spacecraft.
- *Mission Control Centers (MCC)*: their role is to ensure the satellite's mission continues undisturbed by analyzing the telemetry sent by the spacecraft. Telemetry consists of messages transmitted by the satellites containing critical information, such as their exact orbit or internal hardware measurements; this data allows operators to detect anomalies and perform necessary corrective actions.
- *User Segment*: this component comprises all the end-user devices, which receive messages after they are forwarded by the *Ground Stations*. Moreover, devices equipped with the necessary peripherals to communicate directly with the *space segment*, such as ships, airplanes, and Satellite TV, also fall into this category.

2.1.3 Link Segment

This segment is represented by the radio waves that link the Earth to space and the satellites to one another, as well as the hardware components used to establish these connections. Specifically, the *link segment* can be categorized as follows [4, p. 381]:

- *Ground-to-Ground*: connects the *ground stations* to the *user segment* and vice versa.
- *Ground-to-Space*: also known as the *uplink*, it represents the communication path from the *ground segment* to the *space segment*.
- *Space-to-Ground*: also known as the *downlink*, it represents the communication path from the *space segment* to the *ground segment*.
- *Space-to-Space*: connects the spacecraft within the *space segment* to one another, even across different orbital altitudes.

2.2 SATCOM Threats

Due to the distributed architecture of SATCOM networks, security threats targeting any individual segment must be rigorously addressed. Tampering with a single component can have cascading effects, jeopardizing the other segments and compromising both the spacecraft's mission and the broader network to which it is connected. Some threats associated with each segment are outlined below.

2.2.1 Space Segment Threats

While spacecraft are physically out of reach for potential attackers, this physical isolation does not exempt the *space segment* from cyber threats.

Examples include *software attacks*, in which malicious software could be introduced into the satellite's systems via the *link segment*. Alternatively, in a *Satellite-as-a-Service (SataaS)* architecture, an attacker could exploit the shared infrastructure by gaining access to the spacecraft's resources under the guise of a legitimate user. Furthermore, *Denial of Service (DoS)* attacks pose a significant threat to the *space segment*, as an excessive volume of requests sent to the targeted satellite could overwhelm its processing capabilities, temporarily or permanently disrupting its mission.

Beyond software vulnerabilities, *hardware attacks* target the physical infrastructure of the satellite. A particularly insidious example is the *supply chain attack*, which compromises the manufacturing process of the spacecraft by exploiting the decentralized production models typical of the *new space* industry [4, p. 386–389].

Although significantly harder to mount and execute, another critical threat to the *space segment* is posed by *kinetic or anti-satellite (ASAT) weapons*. These systems consist of high-energy beams or physical interceptors that aim to blind the spacecraft's communications or physically destroy its components [7, p. 5677].

Finally, another catastrophic threat is the risk of *satellite hijacking*. Through command spoofing techniques, an attacker could force the spacecraft to deviate from its intended orbit, often leading to the complete destruction of the satellite as it burns up upon atmospheric reentry [4, p. 386–389].

2.2.2 Ground Segment Threats

Compared to the *space segment*, the *ground segment* represents a much more accessible entry point for attackers due to its terrestrial location. Moreover, despite the physical distance between *Ground Stations* and the spacecraft, the impairment of a terrestrial facility can easily jeopardize the satellite itself. For instance, malware contaminating a station on Earth could easily be transferred to the *space segment* via the *link segment*.

Similar to the threats facing the space infrastructure, *software attacks* could lead to the temporary deactivation of the communication channels used to send messages to the *space segment*. If the compromised facility is a *Mission Control Center (MCC)*, operators would be unable to monitor the satellite's telemetry or detect potential anomalies. Consequently, a synchronized cyberattack targeting the *Space Segment* simultaneously would go completely unnoticed.

While *Denial of Service (DoS)* and *Distributed Denial of Service (DDoS)* attacks also pose a significant threat to the *ground segment*, a vulnerability exclusively tied to the terrestrial infrastructure is *Unauthorized Access to Ground Stations*. If physical or logical authentication mechanisms are missing, poorly designed, or vulnerable, an adversary could gain unauthorized entry, ultimately leading to a complete takeover of the spacecraft's control systems [4, p. 389–391].

2.2.3 Link Segment Threats

Lastly, the *link aegment* is arguably the most vulnerable of the three segments, primarily due to its wireless nature. The most straightforward threat is *eavesdropping*, which consists of the unauthorized interception of communications. The adversaries' preferred target for eavesdropping

is typically the *downlink*, as the signal is directed towards Earth over a wide footprint; conversely, the highly directional trajectory of the *uplink* makes it significantly more difficult to capture data [4, p. 393].

Link jamming is another critical threat to consider. Equipped with sufficiently powerful hardware, adversaries can execute a *DoS* attack by flooding the communication channel with noise. Just as with eavesdropping, it is generally easier to jam the *downlink* than the *uplink*. This occurs because the signal arriving from space is inherently weak by the time it reaches Earth, whereas a terrestrial attacker would find it extremely challenging to overpower the highly amplified *uplink* signals transmitted by *Ground Stations*.

Furthermore, *signal spoofing* involves falsifying or replaying messages over the communication link. By doing so, an attacker can deceive the receiving system if robust security and authentication mechanisms are missing or insufficient [4, p. 391–394].

2.2.4 An Emergent Threat: Machine Learning

Although *Artificial Intelligence (AI)* and *Machine Learning (ML)* are widely employed in defensive contexts, such as *Intrusion Detection Systems (IDS)* and *Intrusion Prevention Systems (IPS)*, they also represent a formidable offensive threat.

A first example is linked to the energy-constrained nature of the space environment. Due to strict limits on power and computational resources, onboard ML models dedicated to security are often lightweight and less complex. Consequently, attackers can exploit these limitations by injecting imperceptible adversarial noise to elude the defensive models [7, p. 5682].

More critically for the scope of this study, the increasing accessibility of advanced ML algorithms has revolutionized cryptanalysis. While ML can be used to perform traffic analysis by identifying patterns in encrypted SATCOM communications due to the presence of predictable packet headers, an even more critical risk arises when ML is combined with *Side-Channel Attacks (SCA)*.

Adversaries can leverage ML models to analyze physical leakages, such as the power consumption of the satellite's hardware during encryption. Thanks to the ability of ML to uncover complex, non-linear correlations within noisy data, attackers can profile these leakages and completely recover the secret keys. In the context of shared architectures like *SataaS*, where adversaries might run software on the same physical hardware, ML-based SCAs represent a severe threat if the isolation mechanisms designed to prevent software-based power monitoring are missing or insufficient.

Furthermore, while the traditional execution of SCAs requires placing specific measurement probes on the target hardware, which is an unfeasible task for a satellite already in orbit, a sophisticated *supply chain attack* can circumvent this limitation. Adversaries could compromise key satellite components, such as cryptographic chips or CPUs, by implanting the necessary hardware trojans during the manufacturing process, establishing a covert channel to transmit the leaked information once deployed in space.

Lastly, to compromise the SATCOM infrastructure via an SCA, an adversary could exploit the interconnection between the network segments by targeting the more physically accessible *ground*

stations. Since the ground and space segments must share the same cryptographic parameters to establish a secure link, successfully extracting the secret key from a ground terminal yields the exact same catastrophic result for the entire network.

2.3 Defensive Strategies

To counter the vast range of threats discussed previously, SATCOM security relies on various defensive strategies that can be broadly categorized into two main paradigms. While both approaches aim to secure satellite communications, they achieve this through fundamentally different methodologies.

2.3.1 Physical Layer Security (PLS)

The first approach leverages the intrinsic physical properties of the wireless communication channel to provide countermeasures against specific threats. It minimizes resource consumption while ensuring a high level of security under targeted operational circumstances.

A prominent example of this technique relies on the concept of the *wiretap channel*. By employing specialized beamforming antennas, the sender can focus the transmission beam, confining the signal's footprint exclusively to the legitimate receiver's location. Simultaneously, artificial noise can be injected into the spatial directions outside this main beam to degrade the signal-to-noise ratio for potential eavesdroppers [8, p. 3–4].

A major advantage of *PLS* is that it is significantly less demanding in terms of energy consumption compared to high-level *cryptography* [8, p. 2]. Consequently, this technology is particularly well-suited for SATCOM environments, where spacecraft must operate under strict energy and computational constraints.

However, this defensive strategy presents notable limitations. For instance, PLS mechanisms cannot detect or correct data integrity issues, such as bit-flip errors, which frequently occur during transmission due to long propagation distances and the interference of space radiation. To address these vulnerabilities, current literature frequently explores the combined use of *PLS* and high-level *cryptography*. This hybrid approach aims to achieve comprehensive security by allowing the two methodologies to mutually compensate for each other's limitations.

2.3.2 Traditional Cryptography

Regardless of the transmission medium, *cryptography* represents a fundamental solution for ensuring data confidentiality. By relying on either a single secret key or a public-private key pair, cryptographic algorithms conceal the true content of a message, rendering the transmitted data indistinguishable from random noise.

Algorithms that rely on a single shared key are known as symmetric-key ciphers. For this cryptographic approach to function securely, the secret key must be distributed through a secure channel; the compromise of such critical information leads to a complete breach of communication confidentiality until a new secret key can be established.

To address the key distribution challenge, researchers continuously explore lightweight and efficient methods to establish secure channels. Proposed solutions include *Identity-Based Encryption (IBE)*, where the receiver's public identity serves as the public key while a trusted third party generates the corresponding private key [9, pp. 2193–2194], as well as *Quantum Key Distribution (QKD)*, which leverages quantum mechanics to securely establish a symmetric key [7, p. 5683]. Traditionally, however, SATCOM systems have relied on pre-loading a set of master keys onto the spacecraft prior to its launch, from which dynamically changing session keys are derived [10]. More recently, modern space infrastructures have begun deploying *Elliptic Curve Cryptography (ECC)-based Diffie-Hellman* key exchanges, capitalizing on the computational efficiency and smaller key sizes of ECC [11].

As implied by the adoption of ECC, the resource-constrained nature of the space environment implies the use of highly efficient algorithms to avoid overburdening the satellite's power and computational budget. For this reason, traditional asymmetric-key algorithms like *RSA*, which require substantial computational overhead to process large key pairs, are generally not recommended for SATCOM encryption.

Finally, a major advantage of traditional cryptography over PLS techniques lies in its software-based nature. While deploying a new PLS mechanism requires rigid hardware integration during the satellite's manufacturing phase, cryptographic protocols can potentially be updated or patched remotely, even for spacecraft already in active mission operations.

Furthermore, to combat both eavesdropping and signal spoofing simultaneously, modern SATCOM architectures increasingly rely on Authenticated Encryption with Associated Data (AEAD) schemes, which guarantee both the confidentiality and the integrity of the transmitted information in a single, efficient operational step [3].

2.4 SATCOM Cryptographic State-of-the-Art: AES-256-GCM

Following the directives of the *Consultative Committee for Space Data Systems (CCSDS)*, the most widely adopted cryptographic algorithm in modern satellite infrastructures is the *Advanced Encryption Standard (AES)* operating in *Galois/Counter Mode (GCM)* with a 256-bit key [3].

The rationale behind this choice rests on the intrinsic computational efficiency of AES. While older deployments often relied on 128-bit keys, the escalating threat posed by quantum computing has shifted the security paradigm. Quantum-based attacks, specifically those leveraging Grover's algorithm, can effectively halve the search space of a symmetric-key block cipher, reducing a 128-bit key security level from 2^{128} to a computationally feasible 2^{64} [12]. Conversely, halving the 2^{256} search space of AES-256 still leaves an effective security level of 2^{128}, which remains robustly secure against foreseeable quantum cryptanalysis.

AES is a block cipher that requires its input to be processed in fixed blocks of 128 bits. To adapt this core algorithm to variable-length messages, a specific *mode of operation* must be employed. The CCSDS recommends GCM, an advanced alteration of the classic *Counter Mode (CTR)*. GCM acts as an *Authenticated Encryption with Associated Data (AEAD)* scheme, providing both message confidentiality and authentication simultaneously. This is achieved by generating

an authentication tag derived from the plaintext, the secret key, and the *Additional Authenticated Data (AAD)*.

The adoption of GCM ensures that any data manipulation or bit-flip errors incurred during transmission can be reliably intercepted; if the authentication tag verification fails, the receiver can discard the corrupted payload and request a retransmission.

Finally, to guarantee cryptographic security, AES-GCM requires a strictly unique *Initialization Vector (IV)* associated with each message. Reusing an IV compromises the algorithm's confidentiality. Based on CCSDS recommendations, the optimal IV length is fixed at 96 bits (12 bytes) [12].

Chapter 3

Algorithmic Foundations of AES-256-GCM

As mentioned in the previous chapter, the primary cryptographic standard utilized in modern SATCOM environments is AES-256-GCM.

This chapter provides a comprehensive overview of the mathematical and algorithmic logic underlying this protocol. First, it explores the internal mechanics of the Advanced Encryption Standard (AES) block cipher, strictly following the specifications outlined in the *Federal Information Processing Standard (FIPS) 197* [13]. Subsequently, it details the *Galois/Counter Mode (GCM)* of operation, based on the guidelines established by the *National Institute of Standards and Technology (NIST)* Special Publication 800-38D [14].

By analyzing these standardized primitives, this chapter lays the essential technical groundwork for understanding the specific vulnerabilities targeted by Side-Channel Attacks (SCA), and ultimately justifies the architectural modifications proposed later in this study.

3.1 AES-256

The Advanced Encryption Standard (AES) has established itself as the preeminent symmetric encryption algorithm since the publication of the official *FIPS 197* standard in 2001. Selected by NIST through a rigorous public competition to replace the aging *Data Encryption Standard (DES)*, AES has consistently proven its resilience over the decades. The core algorithm has remained fundamentally unchanged, with the primary adaptation to emerging threats being the transition from a 128-bit to a 256-bit key length.

As a symmetric-key cipher, AES relies on a single shared secret key for both the encryption and decryption processes. This symmetry, combined with the algorithm's computational efficiency, results in a highly lightweight protocol that maintains robust cryptographic security. These characteristics represent the primary reasons for its widespread adoption as the standard for resource-constrained environments like SATCOMs.

To translate these theoretical advantages into a practical experimental framework, this study adopts the open-source `tiny-aes-c` library [2] as the baseline implementation. This C-based library provides a solid foundation for both experimental analysis and proof-of-concept alterations of the standard protocol. Its reliability and minimal computational overhead are well documented, as evidenced by its widespread use as a benchmark reference in recent literature concerning resource-constrained networks, such as Internet of Medical Things (IoMT) [15] and broader IoT

applications [16].

Returning to the theoretical framework, before the actual encryption process begins, the secret key must pass through the KEYEXPANSION routine. This function extends the original 32-byte key into a 240-byte key schedule, which is essential to provide a unique 128-bit round key for the initial key addition phase and for each of the subsequent 14 rounds of the algorithm.

3.1.1 Key Expansion

To comprehend the logic underlying the Key Expansion algorithm, the primary variables and constants are defined as follows:

- N_k: The number of 32-bit *words* comprising the original cipher key. For AES-256, $N_k = 8$.
- N_r: The number of rounds the algorithm will execute. For AES-256, it is fixed at $N_r = 14$.
- w: The linear array of 32-bit words that will hold the complete expanded key schedule.
- `temp`: A temporary 32-bit variable used during transformations.
- `Rcon`: A predefined array of round constant words, used to eliminate symmetries.

The procedure is detailed in the following pseudocode (Algorithm 3.1.1):

```
1:  procedure KEYEXPANSION(key)
2:      i ← 0
3:      while i ≤ N_k − 1 do
4:          w[i] ← key[4 × i, ..., 4 × i + 3]
5:          i ← i + 1
6:      end while                          ▷ When the loop concludes, i = N_k.
7:      while i ≤ (4 × N_r + 3) do
8:          temp ← w[i − 1]
9:          if i mod N_k = 0 then
10:             temp ← SUBWORD(ROTWORD(temp)) ⊕ Rcon[i/N_k]
11:         else if N_k > 6 and i mod N_k = 4 then
12:             temp ← SUBWORD(temp)
13:         end if
14:         w[i] ← w[i − N_k] ⊕ temp
15:         i ← i + 1
16:     end while
17:     return w
18: end procedure
```

In the initial phase (`lines 3-6`), the original secret key is sequentially copied into the first N_k words of the expanded key array w. In the specific case of AES-256 ($N_k = 8$), the array elements $w[0]$ through $w[7]$ will hold the unmodified original key.

The main loop (`lines 7-17`) iterates until the key schedule is completely populated, generating a total of $4 \times (N_r + 1)$ words (i.e., 60 words for AES-256). For each iteration, the word preceding the current index is copied into the `temp` variable (`line 8`). Depending on the index

i, specific non-linear operations may be applied to `temp` to guarantee cryptographic diffusion and security.

Specifically, whenever the index i marks the beginning of a new N_k-word block (i.e., i mod $N_k = 0$), `temp` undergoes a critical three-step transformation (`line 10`). First, RotWord applies a cyclic left permutation to the four bytes within the word, shifting `[a,b,c,d]` to `[b,c,d,a]`. Secondly, the SubWord function substitutes each individual byte using the non-linear AES S-box (which will be explored comprehensively later in this study). Lastly, the result is XORed with the round constant `Rcon` to disrupt any mathematical symmetries.

Furthermore, AES-256 introduces an additional security measure: when $N_k > 6$ (i.e., only for AES-256) and i mod $N_k = 4$, a secondary SubWord transformation is applied to the `temp` variable (`line 12`). This specific step was designed by the algorithm's creators to break potential patterns that could emerge due to the extended 256-bit key length.

Finally, the new word $w[i]$ is generated by computing a bitwise XOR between the heavily transformed `temp` variable and the expanded key word located exactly N_k positions prior (`line 14`).

3.1.2 State Matrix and Round Architecture

AES operates as a block cipher, meaning it processes data in fixed-size blocks of 128 bits (16 bytes). An input block can be represented as an array of 16 bytes:

$$\mathbf{in} = \begin{bmatrix} in_0 & in_1 & in_2 & \dots & in_{15} \end{bmatrix}^T$$

During the execution of the algorithm, this 16-byte input is mapped into a 4×4 matrix of bytes, known as the *State matrix* (S). As defined by the standard, the State matrix is populated column by column:

$$S = \begin{bmatrix} in_0 & in_4 & in_8 & in_{12} \\ in_1 & in_5 & in_9 & in_{13} \\ in_2 & in_6 & in_{10} & in_{14} \\ in_3 & in_7 & in_{11} & in_{15} \end{bmatrix}$$

As a testament to the structural elegance of the algorithm, AES-256 consists of 14 iterations (rounds) of four fundamental transformations applied to the State matrix in a fixed sequence: SubBytes, ShiftRows, MixColumns, and AddRoundKey. The only deviations from this strict regularity are an initial AddRoundKey transformation applied prior to the first round, and the deliberate omission of the MixColumns operation in the final (14th) round.

Finally, each AES transformation, with the exception of AddRoundKey, which is its own mathematical inverse (being a simple XOR operation), possesses a strictly corresponding inverse function utilized during the decryption phase. The decryption algorithm essentially reverses the encryption process by applying these inverse operations (e.g., InvSubBytes, InvShiftRows) in the reverse sequence. Consequently, the 128-bit round keys derived during the Key Expansion phase are also applied in reverse order, starting with the final round key and concluding with the original secret key.

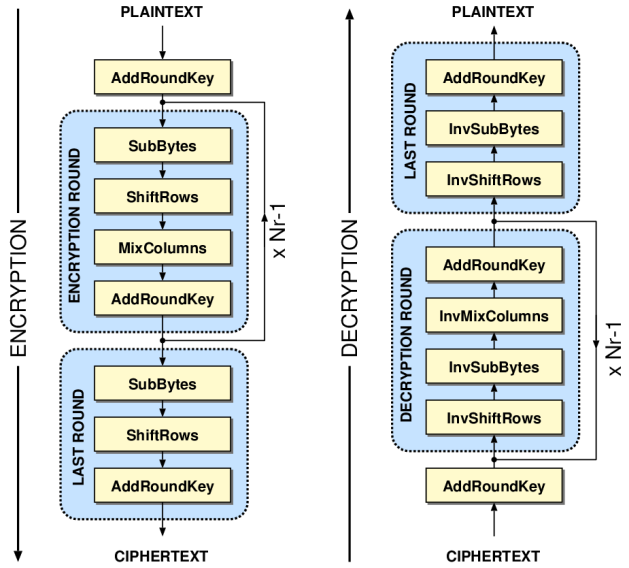


Figure 3.1: Internal structure of the AES algorithm [17].

3.1.3 SubBytes

The SUBBYTES operation is the only strictly non-linear transformation within the AES algorithm. It consists of a byte-by-byte substitution based on a fixed Look-Up Table (LUT) called the *S-Box*. The S-Box is a 16×16 matrix containing every possible byte value, ranging from `0x00` to `0xFF`.

Both the columns and the rows are labeled in hexadecimal format from `0` to `F`. To apply the substitution, each byte of the State matrix is treated as a two-digit hexadecimal value: the first digit determines the row, and the second digit determines the column. The substituted byte is the value located at these intersecting coordinates.

		y															
		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
x	0	63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
	1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
	2	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
	3	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
	4	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
	5	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
	6	d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
	7	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
	8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
	9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
	a	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	79
	b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
	c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
	d	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
	e	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
	f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

Figure 3.2: The AES Substitution Box (S-Box) [18].

For instance, considering the binary input $X = 01101100_2$, its hexadecimal representation is `0x6C`. By intersecting row `6` and column `C` in the S-Box, the output is `0x50`. Therefore,

SUBBYTES(0x6C) = 0x50.

The Construction of the S-Box

The design of the S-Box stems from the fundamental cryptographic necessity to introduce strong non-linearity (confusion) into the cipher, rendering it highly resistant to linear and differential cryptanalysis.

To achieve this, the S-Box is constructed through two mathematical steps. First, each element is mapped to its multiplicative inverse in the finite Galois Field GF(2^8). The multiplicative inverse x^{-1} of a value x in this finite field is defined as the value that, when multiplied by x, yields 1:

$$x \times x^{-1} \equiv 1 \pmod{x^8 + x^4 + x^3 + x + 1} \tag{3.1}$$

The irreducible polynomial $x^8 + x^4 + x^3 + x + 1$ corresponds to the binary sequence 100011011_2. This specific polynomial was selected by the AES creators to ensure the output is strictly constrained within an 8-bit length. By definition, the inverse of 0x00 is mapped to itself.

Secondly, to break algebraic symmetries and prevent structural attacks, each multiplicative inverse b undergoes a specific affine transformation over GF(2^8):

$$b' = (A \times b) \oplus c$$

where A is a fixed 8×8 bit matrix, and c is the constant column vector $[1,1,0,0,0,1,1,0]^T$ (corresponding to 0x63).

Vulnerability of the Software LUT Implementation

Because dynamically calculating the multiplicative inverse in GF(2^8) during every round would introduce an enormous computational overhead, standard software implementations, such as the `tiny-aes-c` library or those deployed on resource-constrained IoT and edge devices, rely entirely on precalculating these transformations and storing them in a static S-Box LUT.

Consequently, the necessity for these constrained devices to repeatedly access the LUT stored in memory or cache generates a deterministic physical energy leakage during each substitution. This power footprint is directly correlated with the specific byte value being processed, creating a vulnerability that can be readily exploited in Power Analysis Side-Channel Attacks.

Conversely, modern high-performance processors mitigate this vulnerability by utilizing dedicated cryptographic hardware extensions (e.g., Intel AES-NI or ARM CE). These hardware modules execute AES transformations directly within the CPU using dedicated logic gates, thereby eliminating the need for memory-based LUT lookups and effectively neutralizing this specific class of side-channel leakage.

3.1.4 ShiftRows

The SHIFTROWS transformation is a linear operation primarily responsible for introducing horizontal diffusion across the State matrix. It adheres to the following mathematical formulation:

$$s'_{r,c} = s_{r,(c+r) \bmod 4} \quad \text{for } 0 \leq r < 4 \text{ and } 0 \leq c < 4 \tag{3.2}$$

Following the application of SHIFTROWS, the resulting 4×4 matrix is altered such that the first row remains completely untouched. The second, third, and fourth rows undergo a cyclic left shift of one, two, and three byte positions, respectively.

$$S = \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} \xrightarrow{\text{SHIFTROWS}} S' = \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,1} & s_{1,2} & s_{1,3} & s_{1,0} \\ s_{2,2} & s_{2,3} & s_{2,0} & s_{2,1} \\ s_{3,3} & s_{3,0} & s_{3,1} & s_{3,2} \end{bmatrix}$$

While mechanically straightforward, this transformation is cryptographically vital. By shifting the rows, it ensures that the four bytes originally comprising a single column are distributed evenly across four distinct columns. Consequently, when the subsequent MIXCOLUMNS transformation is applied, it will mathematically mix bytes from different original columns, guaranteeing a rapid and robust propagation of the avalanche effect throughout the entire State.

3.1.5 MixColumns

Complementing the horizontal diffusion introduced by SHIFTROWS, the MIXCOLUMNS transformation is the primary operation responsible for applying vertical diffusion across the State matrix. This function ensures that each output byte in a column is mathematically dependent on every input byte within that same column.

To achieve this, each column is treated as a four-term polynomial and multiplied by a fixed matrix over the Galois Field GF(2^8):

$$\begin{bmatrix} s'_{0,c} \\ s'_{1,c} \\ s'_{2,c} \\ s'_{3,c} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{bmatrix} \quad \text{for } 0 \leq c < 4 \tag{3.3}$$

By expanding the matrix multiplication, where $\bullet$ denotes multiplication in GF(2^8) and $\oplus$ denotes bitwise XOR (addition in GF(2^8)), the individual bytes of the new column are calculated as follows:

$$\begin{aligned} s'_{0,c} &= (02 \bullet s_{0,c}) \oplus (03 \bullet s_{1,c}) \oplus s_{2,c} \oplus s_{3,c} \\ s'_{1,c} &= s_{0,c} \oplus (02 \bullet s_{1,c}) \oplus (03 \bullet s_{2,c}) \oplus s_{3,c} \\ s'_{2,c} &= s_{0,c} \oplus s_{1,c} \oplus (02 \bullet s_{2,c}) \oplus (03 \bullet s_{3,c}) \\ s'_{3,c} &= (03 \bullet s_{0,c}) \oplus s_{1,c} \oplus s_{2,c} \oplus (02 \bullet s_{3,c}) \end{aligned} \tag{3.4}$$

Software Implementation of GF(2^8) Multiplication: xtime macro

From a practical engineering perspective, performing heavy polynomial multiplications dynamically during every round would be computationally prohibitive for embedded systems. However,

the matrix coefficients were deliberately selected by the AES designers to be exceptionally small (`01`, `02`, and `03`) to allow for radical software optimization.

In lightweight C implementations such as `tiny-aes-c`, multiplication by `02` in $\mathrm{GF}(2^8)$ is elegantly implemented using a macro function typically named `xtime`. This operation consists of a simple bitwise left shift of the byte, followed by a conditional XOR with the irreducible polynomial `0x1B` if the most significant bit prior to the shift was 1.

To rigorously verify this optimization, the generic input byte must be represented as a polynomial:

$$A(x) = a_7x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$$

where the coefficients $a_7, \ldots, a_0$ are individual bits, with a_7 being the Most Significant Bit (MSB).

The hexadecimal value `0x02` corresponds to the binary string `00000010`, which is equivalent to the polynomial x. Therefore, multiplying $A(x)$ by `0x02` yields:

$$A(x) \times x = a_7x^8 + a_6x^7 + a_5x^6 + a_4x^5 + a_3x^4 + a_2x^3 + a_1x^2 + a_0x$$

In computer architecture, multiplying by x corresponds exactly to a logical bitwise left shift of one position. However, to ensure the result remains within the finite field $\mathrm{GF}(2^8)$, the maximum degree of the resulting polynomial must not exceed x^7. This constraint presents two possible cases:

- $a_7 = 0$**:** After the multiplication, the coefficient of x^8 is 0. The highest degree remains x^7, and no further mathematical reduction is necessary.
- $a_7 = 1$**:** The resulting polynomial contains an x^8 term, thus falling outside the finite field. To bring the result back into $\mathrm{GF}(2^8)$, it must be divided by the AES irreducible polynomial ($x^8 + x^4 + x^3 + x + 1$), and replaced by the remainder of this division. In binary arithmetic over $\mathrm{GF}(2)$, this modulo reduction corresponds precisely to an XOR operation with the irreducible polynomial:

$$(A(x) \times x) \oplus (x^8 + x^4 + x^3 + x + 1)$$

Building on this logic, multiplication by `03` is effortlessly computed. Since `0x03` corresponds to the binary `00000011` (the polynomial $x+1$), the multiplication $A(x) \times$ `0x03` expands to:

$$A(x) \times (x+1) = (A(x) \times x) + (A(x) \times 1) = (A(x) \times x) + A(x)$$

The first term, $(A(x) \times x)$, is exactly the `02` multiplication (`xtime`) just demonstrated. Meanwhile, polynomial addition in $\mathrm{GF}(2^8)$ is mathematically equivalent to a simple bitwise XOR. Thus, the operation seamlessly translates to $(\texttt{02} \bullet A) \oplus A$.

This architectural design guarantees that the MIXCOLUMNS transformation remains exceptionally fast and highly efficient, relying strictly on basic CPU bitwise operations without necessitating additional memory-heavy LUTs.

3.1.6 AddRoundKey

The ADDROUNDKEY transformation is the first operation applied in the algorithm. It consists of a straightforward bitwise XOR operation between the current *State matrix* and the four words of the *expanded key* corresponding to the current round.

For a given column c (where $0 \leq c < 4$) and a specific round, the transformation is defined as:

$$\begin{bmatrix} s'_{0,c} \\ s'_{1,c} \\ s'_{2,c} \\ s'_{3,c} \end{bmatrix} = \begin{bmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{bmatrix} \oplus \begin{bmatrix} w_{4\times\text{round}+c} \end{bmatrix}$$

Consider, for instance, the application of the ADDROUNDKEY transformation to the first column of the state ($c = 0$) during the first round. Let the initial state column and the corresponding word from the expanded key array (w_4) be defined as follows:

$$\begin{bmatrix} s_{0,0} \\ s_{1,0} \\ s_{2,0} \\ s_{3,0} \end{bmatrix} = \begin{bmatrix} \texttt{04} \\ \texttt{E0} \\ \texttt{48} \\ \texttt{28} \end{bmatrix} \quad \text{and} \quad w_{4\times1+0} = w_4 = \begin{bmatrix} \texttt{D6} \\ \texttt{AA} \\ \texttt{74} \\ \texttt{FD} \end{bmatrix}$$

By applying the bitwise XOR ($\oplus$) between the two column vectors, the resulting updated state column is computed as:

$$\begin{bmatrix} s'_{0,0} \\ s'_{1,0} \\ s'_{2,0} \\ s'_{3,0} \end{bmatrix} = \begin{bmatrix} \texttt{04} \\ \texttt{E0} \\ \texttt{48} \\ \texttt{28} \end{bmatrix} \oplus \begin{bmatrix} \texttt{D6} \\ \texttt{AA} \\ \texttt{74} \\ \texttt{FD} \end{bmatrix} = \begin{bmatrix} \texttt{D2} \\ \texttt{4A} \\ \texttt{3C} \\ \texttt{D5} \end{bmatrix} \tag{3.5}$$

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